

Siddharth Somasundaram

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CURRENT APPOINTMENT

Graduate Research Assistant

Massachusetts Institute of Technology

Advisor: Ramesh Raskar

EDUCATION

Massachusetts Institute of Technology

PhD, Media Lab

Cambridge, MA

2024 – Present

Massachusetts Institute of Technology

MS, Media Lab

Cambridge, MA

2022 – 2024

University of California, Los Angeles

BS, Electrical Engineering

Los Angeles, CA

2017 – 2021

HONORS AND AWARDS

NSF Graduate Research Fellowship

2024

CVPR Best Paper Finalist

2024

Outstanding Bachelors in ECE Finalist

2021

Eta Kappa Nu

2019

Dan and Helen Low Scholarship in Engineering

2019

UCLA ECE Fast Track Program

2017

UCLA Regent's Scholarship Finalist

2017

INTERNSHIPS

Serve Robotics

Autonomy Team Intern

Manager: Abhishek Bhatia

Mountain View, CA

2026

University of Toronto

Visiting Graduate Student

Host: Kyros Kutulakos and David Lindell

Toronto, Canada

2024

MIT Media Lab, Camera Culture

Research Staff

Advisor: Ramesh Raskar

Cambridge, MA

2021 – 2022

HRL Laboratories

Quantum Optics Research Intern

Manager: Thaddeus Ladd

Malibu, CA

2020

The Aerospace Corporation

Photonics Technology Engineer Intern

Manager: William Lotshaw

El Segundo, CA

2019

JOURNAL PUBLICATIONS

* EQUAL CONTRIBUTION

Please refer to my [Google Scholar](#) for a complete list.

Imaging Hidden Objects with Consumer LiDAR via Motion Induced Sampling

2026

S. Somasundaram, A. Young, A. Dave, A. Pediredla, R. Raskar

Nature (to appear)

Event Cameras Meet SPADs for High-Speed, Low-Bandwidth Imaging

2025

M. Muglikar, S. Somasundaram, A. Dave, E. Charbon, R. Raskar, D. Scaramuzza

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

A Roadmap for Generative Design of Visual Intelligence

2025

K. Tiwary, T. Klinghoffer, A. Young, S. Somasundaram, N. Behari, A. Dave, B. Cheung,
D. Nilsson, T. Poggio, R. Raskar
MIT Press: Impacts of Generative AI

Detection and Mapping of Specular Surfaces Using Multibounce Lidar Returns 2023
C. Henley, S. Somasundaram, J. Hollmann, R. Raskar
Optics Express

Room-Temperature Mid-Wavelength Infrared InAsSb Nanowire Photodetector Arrays with Al₂O₃ Passivation 2019
D. Ren, Z. Rong, K. Azizur-Rahman, S. Somasundaram, M. Shahili, D. Huffaker
Nano Letters

Feasibility of Achieving High Detectivity at Short- And Mid-Wavelength Infrared using Nanowire Photodetectors with P-N Heterojunctions 2019
D. Ren, Z. Rong, K. Azizur-Rahman, S. Somasundaram, M. Shahili, D. Huffaker
Nanotechnology

A Three-Dimensional Insight into Correlation Between Carrier Lifetime And Surface Recombination Velocity for Nanowires 2018
D. Ren, Z. Rong, S. Somasundaram, K. Azizur-Rahman, B. Liang, D. Huffaker
Nanotechnology

Uncooled Photodetector at Short-Wavelength Infrared Using InAs Nanowire Photoabsorbers on InP with P-N Heterojunctions 2018
D. Ren, X. Meng, Z. Rong, C. Minh, A. C. Farrell, S. Somasundaram,
K.M. Azizur-Rahman, B.S. Williams, D.L. Huffaker
Nano Letters

CONFERENCE PUBLICATIONS

* EQUAL CONTRIBUTION

Please refer to my [Google Scholar](#) for a complete list.

Spatiotemporal Flux Probing for Single-Photon Videography 2026
J. Yan*, M. Forlivesi*, B. Tan, A. Xie, S. Somasundaram, S. Nousias
European Conference on Computer Vision (ECCV)

Shoot-Bounce-3D: Single-Shot Occlusion-Aware 3D from Lidar by Decomposing Two-Bounce Light 2025
T. Klinghoffer, S. Somasundaram*, X. Xiang*, Y. Fan, C. Richardt, A. Dave, R. Raskar, R. Ranjan
SIGGRAPH Asia

Blurred LiDAR for Sharper 3D: Robust Handheld 3D Scanning with Diffuse LiDAR and RGB 2025
N. Behari, A. Young, S. Somasundaram, T. Klinghoffer, A. Dave, R. Raskar
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
Highlight (Top 2.98%, 13,008 submissions)

Handheld Mapping of Specular Surfaces using Consumer-Grade Flash LiDAR 2024
T-H. Lin, C. Henley, S. Somasundaram, A. Dave, M. Laifenfeld, R. Raskar
IEEE International Conference on Computational Photography (ICCP)

PlatoNeRF: 3D Reconstruction in Plato's Cave via Single-View Two-Bounce Lidar 2024
T. Klinghoffer, X. Xiang*, S. Somasundaram*, Y. Fan, C. Richardt, R. Raskar, R. Ranjan
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
Oral, Best Paper Finalist (Top 0.79%, 11532 submissions)

Role of Transients in Two-Bounce Non-Line-of-Sight Imaging 2023
S. Somasundaram, A. Dave, C. Henley, A. Veeraraghavan, R. Raskar
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
ICCP Spotlight Poster (12 of 79 submissions)

Physics vs. Learned Priors: Rethinking Camera and Algorithm Design for Task-Specific Imaging 2022
T. Klinghoffer*, S. Somasundaram*, K. Tiwary*, R. Raskar
IEEE International Conference on Computational Photography (ICCP)

INVITED TALKS

Serve Robotics <i>Dual-Return LiDAR for Collision Avoidance</i>	2026
Imagination in Action <i>What happens when your smartphone can see around corners?</i> Host: John Werner	2026
Purdue Computational Imaging Group <i>Repurposing Consumer LiDAR for Extreme Vision</i> Host: Sotiris Nousias	2026
AI + X Global Talent Community Exchange <i>Seeing the Invisible with Everyday Cameras</i> MIT Museum	2025
New England Computer Vision Workshop <i>3D Reconstruction of Occluded and Specular Objects using Multi-Bounce LiDAR</i> Dartmouth College	2023
IIT Madras <i>Shadows in Space-Time for Non-Line-of-Sight Imaging</i> Host: Kaushik Mitra	2023
CMU Reading Group <i>Role of Transients in Two-Bounce Non-Line-of-Sight Imaging</i> Host: Matthew O'Toole	2022

PUBLIC DEMONSTRATIONS

- [P.2] **S. Somasundaram**, A. Young, N. Tsao, A. Dave, A. Pediredla, R. Raskar, “Real-Time Non-Line-of-Sight Tracking with Low-Cost Sensors”, *ICCP Demos 2025*
- [P.1] A. Young, **S. Somasundaram**, N. Tsao, A. Dave, A. Pediredla, R. Raskar, “Real-Time Non-Line-of-Sight Tracking with Low-Cost Sensors”, *CVPR Demos 2025*

REFeree SERVICE

NeurIPS	2026–
SIGGRAPH Asia	2026–
ECCV	2026–
SIGGRAPH	2026–
ACM Transactions on Graphics	2026–
IEEE Transactions on Computational Imaging	2024–
ICIP	2026–

THESES

Mobile Multi-Bounce LiDAR M.S. Thesis, <i>Massachusetts Institute of Technology</i>	2024
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MEDIA COVERAGE

MIT News PlatoNeRF: 3D Reconstruction in Plato’s Cave via Single-View Two-Bounce Lidar [web]	2023
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